

A Methodology for Estimating Emergency Evacuation Times

STEPHEN W. TWEEDIE
JAMES R. ROWLAND
STEPHEN J. WALSH
RONALD P. RHOTEN
Oklahoma State University
PAUL I. HAGLE
Fort Worth Texas

A methodology based on probabilistic mobilization time curves and pertinent evacuation network details is described for estimating the times required for the partial or total evacuation of an emergency planning zone (EPZ). Such an EPZ evacuation study is required by the U.S. Nuclear Regulatory Commission (NRC) for existing and proposed nuclear power operating plants. Inputs and outputs for the resulting computer model developed from the methodology are discussed for a nuclear power plant licensing application by Public Service Company of Oklahoma. With typical input conditions for the example under consideration, the computer model predicted an average total evacuation time of approximately two and a half hours, excluding notification and confirmation times.

Following the Three Mile Island (TMI) accident, the U.S. Nuclear Regulatory Commission (NRC) now requires that estimates be provided for all existing and proposed nuclear power plant facilities predicting the times required for partial and total evacuation of the population within a 10-mile radius of the facility, an area referred to as the plume exposure Emergency Planning Zone (EPZ).¹ Partial evacuations involve combinations of 90-degree sectors and 2-, 5-, and 10-mile radii.² In brief, the NRC requires 88 separate evacuation time estimates based on 10 different areal configurations in both normal and adverse weather conditions, and 4 "incident time of occurrence" (ITO) cases.

Previously, Hans and Sell evaluated the risks of evacuation by summarizing data for 54 incidents of all types involving the evacuation of over 1.1 million people during the period from 1960 to 1973.³ In January 1980, Hubenette et al. determined evacuation time estimates for areas near the Rancho Seco Power Plant in California.⁴ Reported the same month were evacuation time estimates for the EPZ surrounding the Trojan nuclear power plant located near the boundary of Oregon and Washington.⁵ Current guidelines for determining evacuation time estimates set up general performance specifications that indicate a need for traffic network details based on the Highway Capacity Manual.⁶ Urbanick et al. suggested some basic techniques for transportation planning in June 1980, and described the state-of-the-art in estimating EPZ evacuation times.⁷ Sheffi, Mahmassani, and Powell, in 1981, developed the NETVACI (a simulation model that describes the traffic pattern in a transportation network during an emergency evacuation), and applied it to determine emergency evacuation times around four nuclear plants.⁸ Kahn developed a local area traffic model (TRAFIC), which is designed to provide three basic transportation planning functions: (1) trip generation, (2) trip distribution, and (3) traffic assignment.⁹ More recently, Moeller, Urbanick, and Derosiers developed the Calculate Logical Evacuation and Response (CLEAR) model for estimating evacuation times with features including a random starting time and a random loading pattern on roadways.¹⁰

Prior to the 1979 accident at TMI, formal planning for protective actions including evacuation time was generally not required beyond the low-population zone (LPZ). The LPZ is defined as "an area containing residents, the total number and density of which are such that there is a reasonable probability that appropriate protective measures can be taken on their behalf in the event of a serious accident."¹¹ In 1979, the NRC-EPA task force recommended that a 10-mile radius Emergency Planning Zone (EPZ) be defined around each nuclear station, and to be termed the *Plume Exposure Zone*. Within the 10-mile radius planning zone, the task force suggested that shelter and/or evacuation would likely be the immediate protective actions recommended for the general public. Evacuation is especially recommended for the population within 5 miles of a reactor.¹²

Zeigler, Brunn, and Johnson identified the spatial and temporal parameters of evacuation behavior among residents surrounding the Three Mile Island nuclear power plant.¹³ Their research provides a conceptual model of evacuation decision making in response to a nuclear emergency. They indicate that voluntary evacuation following the emergency at TMI clearly reveals a distance-decay relationship that illustrates the evacuation shadow phenomena (tendency of an official evacuation advisory to cause departure from a much larger area than was originally intended) and the effect of government evacuation directives.¹⁴

The TMI experience indicated the need for contingency plans including evacuations based upon different kinds of accidents, with different time scales and different radiation exposures.¹⁵ "Societies using nuclear power today must accept major accidents not only as a theoretical possibility of no practical consequence, but as a risk to include in actual planning."¹⁶

This article describes a methodology, developed concurrently but independently of the NETVACI and CLEAR models, for estimating emergency evacuation times around a proposed nuclear power plant. As with both of the other models, the methodology presented here determines travel times over local access roads and primary network

routes having traffic congestion. Unlike these previous models that determine typical evacuation time estimates, the methodology here uses statistical analysis with repeated Monte Carlo runs to provide both average results and expected variations in these evacuation time estimates.

The evacuation planning criteria and assumptions contained within this model are inherent within the existing NRC federal emergency planning guidelines. Therefore, evacuation methodology designed within this model is restricted to the defined NRC guidelines, which include, for example, statements regarding evacuation planning regions and population notification assumptions. The NRC evacuation planning guidelines regarding mandatory evacuation planning must be adhered to for nuclear power plant licensing.

DESCRIPTION AND FORMULATION

The case study used to illustrate the methodology is the Public Service of Oklahoma (PSO) proposed Black Fox Station (BFS), located about 23 miles east of Tulsa in a gently rolling plains region containing scattered farmsteads and generally low population densities. While there is only one incorporated town in the EPZ (Inola, Oklahoma: 1980 population 1,650), the western portion of the EPZ includes the fringe of suburban Broken Arrow, a city that is part of the expanding Tulsa metropolitan area.

Information to estimate evacuation time includes a complete description of the population at risk and the transportation systems available. The database development began with a "windshield" survey of the 10-mile radius area, wherein every building was located by utility company employees. The buildings were entered on a datasheet and included information about building use (residence, business, school, etc.), number of parking spaces, and other pertinent descriptors. Information concerning the road network was collected at this same time, with all roads described as to number of lanes, surface type, traffic controls, and bridge types. A set of black-and-white aerial photographs of the entire area was taken at a scale of approximately 1 inch to 400 feet (1:4800), and was used to verify residential and business locations and to update road maps. The population for 1980 was projected to the year 2020 for all population types (existing residential, new residents expected from the growth of the nearby Tulsa metropolitan area, the nuclear plant workforce, population growth due to the economic impact of the plant, institutionalized persons, and visitors to recreation sites).

A 24×24 mile square area was centered on the BFS site in order to accommodate the 10-mile radius planning region around the proposed nuclear power station as specified in NRC evacuation guidelines. This area was divided into square cells of one-half-mile sides, creating a 48×48 cell grid or 2,304 basic area data cells. Because of the prevalence of the Federal Rectangular Survey system in the area, these population cells, oriented in cardinal directions, correspond to quarter-sections with each cell bounded on two adjacent sides by a section line and usually a section line road. Estimates of populations were made for four ITO cases (nighttime, worktime, Sunday morning, and summer Saturday) and for each of the six population types.

For partial evacuation cases, 90-degree sectors were oriented geographically to place boundaries through zones of low population density (see Figure 1). As examples, the Inola community falls with the North sector (N10), and the Broken Arrow suburban area is contained within the West sector (W10).

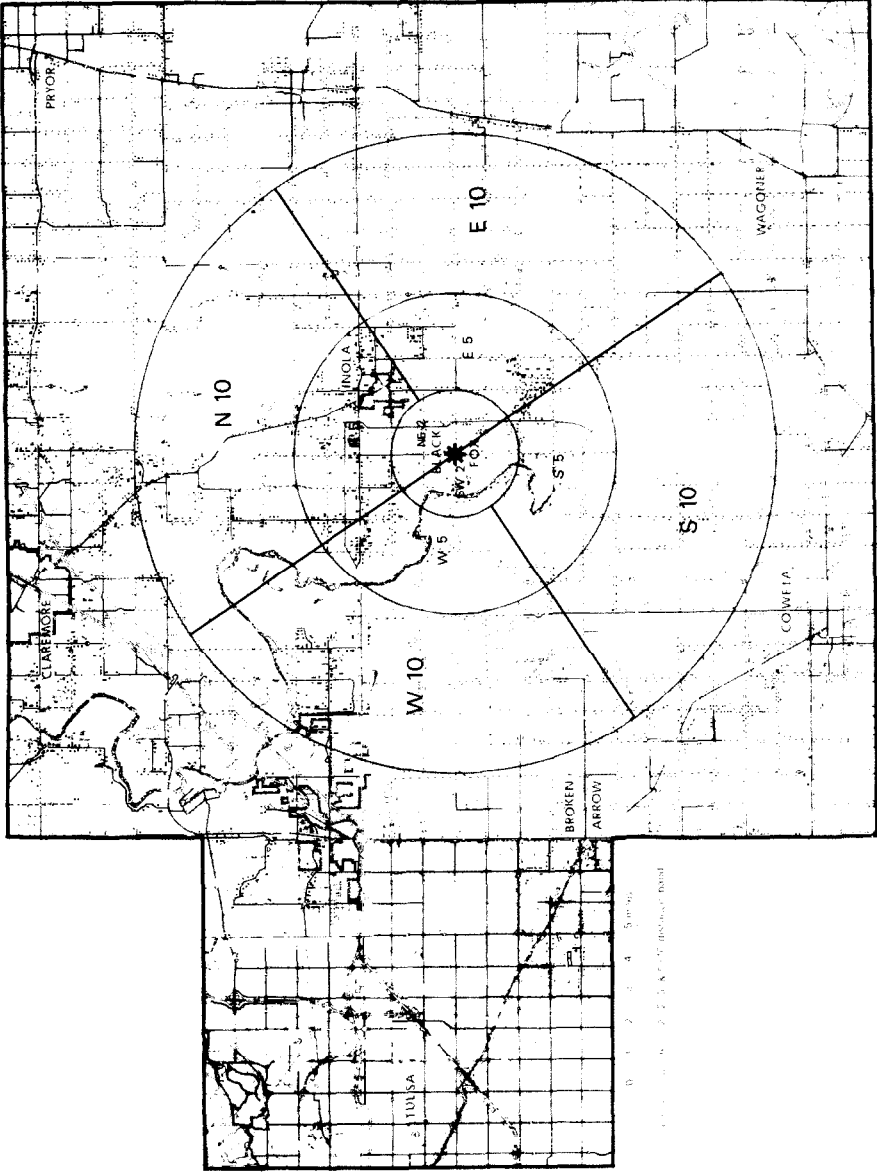


Figure 1. Evacuation Quadrants and Population Sectors for Black Fox Station.

Walsh et al. describe the development of estimated evacuation times completed for the BFS. The estimated evacuation time was completed in three major stages: (1) generation of an appropriate database, (2) prediction of pertinent variables over the expected 40-year life of the nuclear station, and (3) development of a computer simulation program that modeled the evacuation process.¹⁶

The problem was to formulate a methodology based on this gridded structure that would estimate emergency evacuation times by taking into account details of the evacuation network itself.

EVACUATION NETWORK ORGANIZATION

Each cell of the gridded EPZ was viewed as a point of origin for evacuating population with exit routes determined for every one-quarter-square-mile cell. This step involves a number of assumptions about highway quality, capacity, speed, traffic control, and individual decisions. These detailed assumptions could easily be modified, if deemed questionable, without altering the methodology.

For most cells there was a clear "best" route based on (1) travel away from the nuclear site, (2) by the shortest route, and (3) on the best quality road. Although arbitrary decisions were necessary in some cases, such assignments were made with a "worst case" rationale by assigning all of them to a logical but probably most congested route. The model could be modified to allow a selection of alternative routes in a probabilistic fashion. However, it should be recognized that irrational human behavior programmed as a random input may unduly complicate a computer model with only marginal gains in predictability. Further studies are needed to determine the appropriate level of model sophistication.

Three types of nodes on the evacuation network shown in Figure 2 were defined as having special functions in the evacuation process, with some nodes having more than one functional role.

1. *Collector nodes.* Each individual cell was assigned an exit route linking it with an intersection where individual vehicles would be able to enter the traffic flow of a major evacuation route. Collector nodes outside the plume exposure EPZ were identified as Nodes 1 through 26; those inside were identified as Nodes 27 through 56 in Figure 2. The number of vehicles using each of these nodes under the 1980 nighttime population distribution is listed in Table 1.
2. *Exit nodes.* On any given evacuation route, the first recognized node that is outside the plume exposure EPZ was called an *exit node*, identified as Nodes 1 through 26 (see Figure 2). Each exit node is located at an intersection on a major evacuation route. All exit nodes also serve as collector nodes for their own localized area because some vehicles travel directly to them and enter a major highway at that point. Some exit nodes, such as Node 1 on SH 33 West, also serve as important traffic nodes since vehicles from other collector nodes (defined in [3.]) must travel through them. The number of vehicles collecting at each of these nodes is given in Table 1. The total number of vehicles exiting through each of these nodes is also recorded in Table 1.

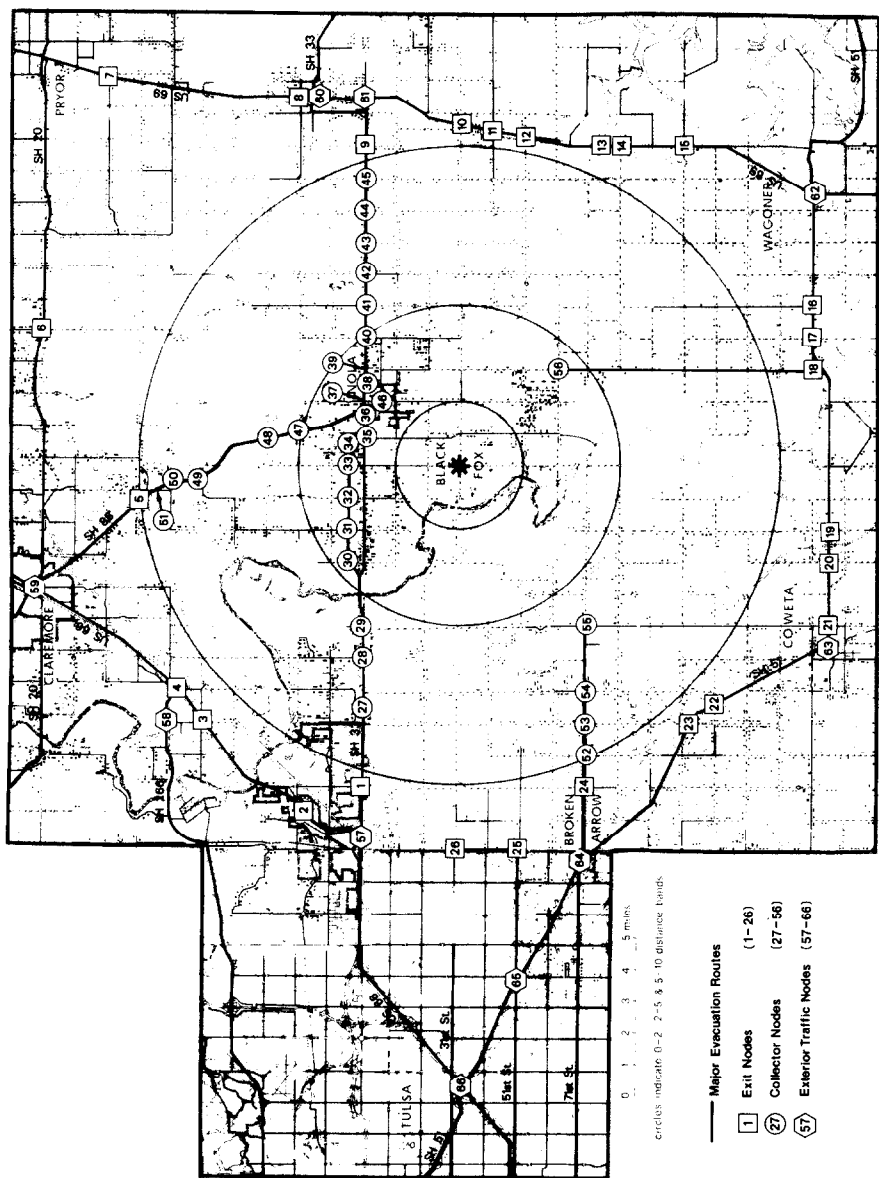


Figure 2. Highway Evacuation Network for Black Fox Station Showing Collector, Exit, and Traffic Nodes.

Table 1. Information for Collector and Exit Nodes: 1980 Nighttime Case.

<i>Traffic Nodes Number</i>	<i>Collector Nodes (Vehicles Entering Evacuation Route)</i>	<i>Exit Nodes (Vehicles Leaving EPZ)</i>	<i>Traffic Nodes Number</i>	<i>Collector Nodes (Vehicles Entering Evacuation Route)</i>
1	0	1537	29	69
2	41	41	30	31
3	6	6	31	108
4	104	104	32	60
5	115	416	33	55
6	9	9	34	87
7	27	27	35	206
8	15	15	36	318
9	9	358	37	61
10	14	14	38	25
11	16	16	39	19
12	31	31	40	139
13	7	7	41	25
14	18	18	42	29
15	11	11	43	13
16	19	19	44	14
17	8	8	45	24
18	43	256	46	145
19	142	142	47	54
20	1	1	48	9
21	40	40	49	72
22	22	22	50	7
23	36	36	51	14
24	4	1272	52	424
25	95	95	53	428
26	9	9	54	363
27	484		55	53
28	119		56	213

3. *Traffic nodes.* Exit nodes and collector nodes are located at intersections on major evacuation routes to enable the model to simulate traffic flow conditions and predict possible points of congestion associated with those key intersections. Thus, both exit nodes and collector nodes also serve as traffic nodes, and traffic flow conditions through those nodes are to be monitored. Beyond the exit nodes, 10 additional exterior intersection nodes (Nodes 57 through 66 in Figure 2) were identified as potential traffic bottlenecks, possibly affecting exiting vehicles. These are included in the methodology to determine if traffic congestion at these nodes could cause traffic to back up into the plume exposure EPZ, thus increasing evacuation time.

The evacuation for each population cell is divided into two parts: (1) travel time over local access roads to reach its collector node, and (2) travel time on primary network roads, which is the time required to exit the EPZ from its collector node.

METHODOLOGY AND COMPUTER MODEL CONSTRUCTION

Under NRC policy guidelines, the total time required for evacuation of an EPZ had four main components:

1. Notification time (time needed to notify all population to be evacuated after an emergency evacuation decisions has been made);
2. Mobilization time (time span between notification of the population and their departure—this is sometimes called *preparation time*);
3. Travel time (following departure from their location when notified—this is the time actually expended in moving out of the EPZ); and
4. Confirmation time (time required to confirm that the population at risk has been evacuated).

The notification time was viewed as a separate problem and is not considered further in this article. Confirmation time is also considered separately once the actual evacuation is completed or near completion. Thus, for purposes of this methodology, the components of evacuation time are estimates of mobilization time (TM), travel time over local access roads (TLT), and travel time on primary network roads (TPT). The total evacuation time for a given vehicle ($TTOT$) is given by:

$$TTOT = TM + TLT + TPT.$$

Sheffi et al. treat travel time through the network (clearance time) as a single component, and their NETVACI model apparently uses a graph representation of the entire road network (see note 8). Such an approach may be more appropriate for an area with a dendritic road pattern than one gridded by section line roads where the number of links, intersections, and potential exit routes expands rapidly. The local access road component of time (TLT) in the equation above simplifies the network flow analysis by eliminating a large number of links and nodes that would be used by very few vehicles.

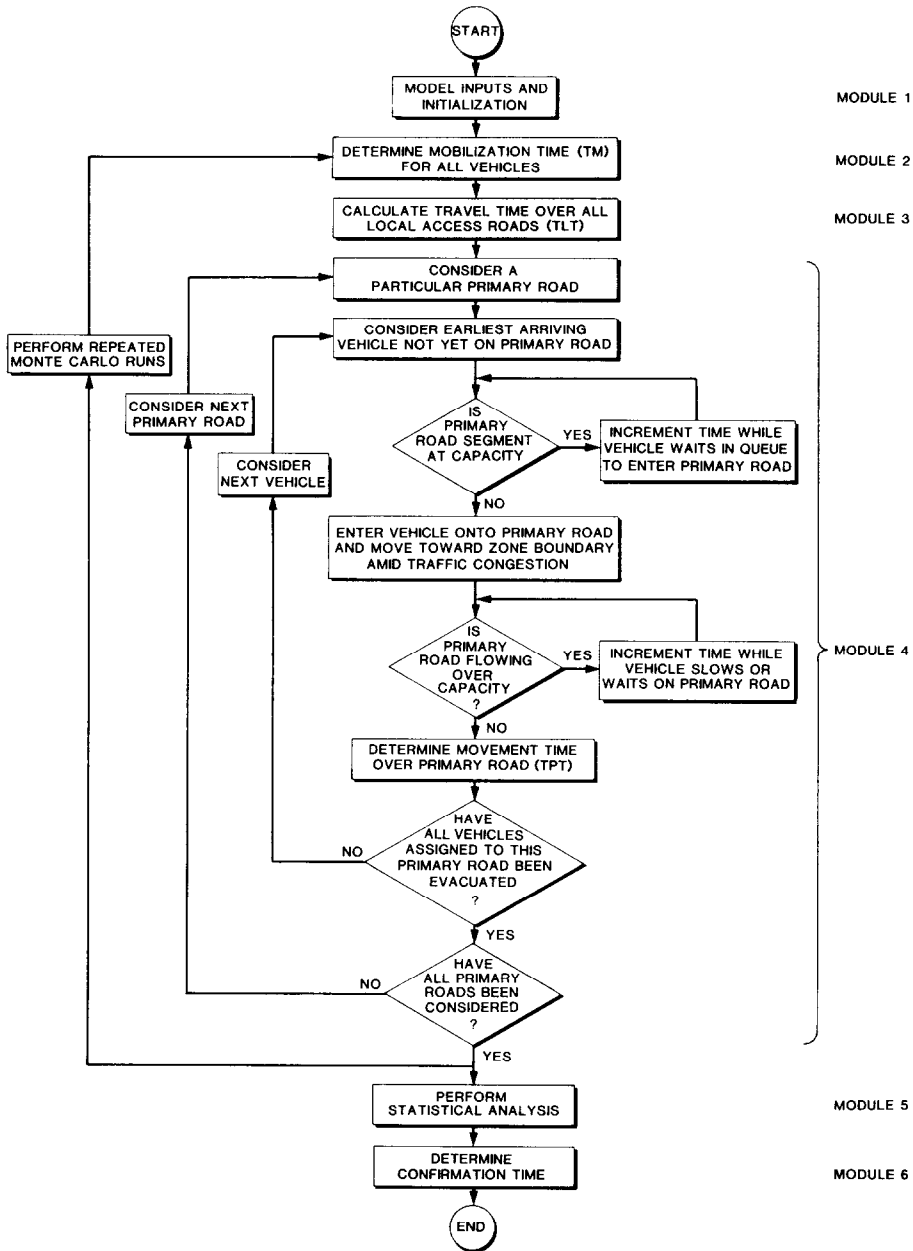
A flowchart of the computer model developed for the evacuation time estimate methodology is shown in Figure 3. The 5 modules of the model perform the following functions:

1. Model inputs and initialization;
2. Determine mobilization time (TM);
3. Calculate travel time over local access roads (TLT);
4. Determine movement time over primary network routes (TPT); and
5. Perform statistical analysis.

Each of these modules is determined in detail in the remainder of this section.

Model Inputs and Initialization

Module 1 of the computer model performs the initialization of the program, including the insertion of all physical data describing the EPZ boundaries, the population in each cell, the primary road network segments in terms of lengths (miles), capacities (vehicles



per hour), and densities (vehicles per mile); and all local access roads with their lengths, capacities, densities, and conditions (paved, gravel, or dirt). In addition, Module 1 indicates the "incident time of occurrence" (*ITO*) and whether normal or adverse weather conditions are in effect.

The capacity (vehicles per hour) and density (vehicles per mile) of each highway segment were determined from its length, number of lanes, speed of traffic, and headway, based on the Highway Capacity Manual (see note 6). The present example assumes a speed of 30 mph for maximum capacity under normal weather conditions, and 15 mph under adverse weather conditions. Other traffic speeds can be selected by the user as desired. The complex interrelationships between speed, capacity, and density are included in the model. As speeds are reduced due to congestion, the capacity of each road segment is reduced accordingly, and its density is increased.

Procedure for Determining Mobilization Time

Previous evacuation studies proved to be of only minimal aid in estimating mobilization time parameters due to their partial and highly tentative nature (see notes 1 and 2). In the absence of an appropriate empirical precedent, mobilization time parameters were determined from information obtained during several meetings with key experts within the state Civil Defense Office. In particular, these experts were questioned to determine the specific amounts of time for which given percentages of the evacuating population could normally be expected to be mobilized. Figure 4 shows a

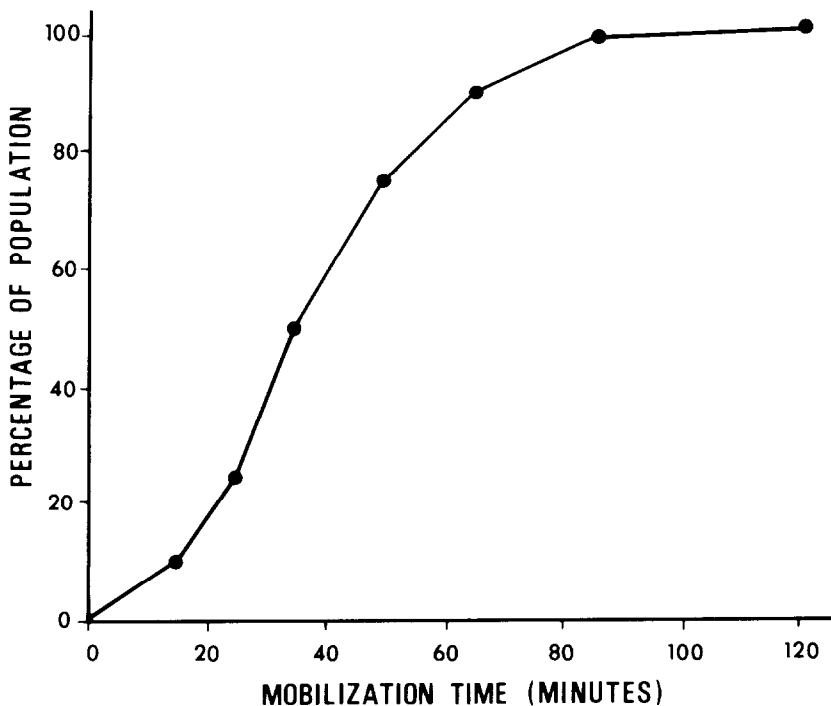


Figure 4. Data for Mobilization Time.

plot of these data, which were then utilized in Module 2 of the model to form a probability distribution function for mobilization time. The data in Figure 4 can be approximated by the Rayleigh probability distribution function given by:

$$F(TM) = 1 - \exp(-TM)^2/1800),$$

where TM is the mobilization time in minutes. Other parameters that correspond to different mobilization time assumptions could be selected for cases involving other EPZ evacuations.

Modeling Movement and Movement Times

In Module 3, capacities, speed of travel, and lengths of local access roads from Module 1 are used to calculate the travel time over local access roads (TLT). The sum of TM and TLT is formed and arranged in ascending order for all vehicles assigned to a particular node for entrance into the primary roadway network. A vehicle that has arrived at a node on the primary roadway network is either allowed to enter on the primary road immediately, or, if the segment is filled to capacity, wait until the congested traffic has cleared enough to provide a space for another vehicle. Time is incremented while the vehicle waits in the queue, thus adding to the time of travel along the primary road (TPT). When a space is available, the vehicle enters the primary road and travels amid congested traffic along this road toward the exit node at the zone boundary. Time is also incremented for vehicles already on a primary exit route that cannot continue into a congested segment, thereby increasing their time of travel on primary routes (TPT). Vehicles approaching an intersection along a main highway have priority over those entering from a side road.

The two alternating steps in the process of load and flow occur during a user-specified time interval to produce an effective “pulsating” operation in terms of calculations performed for this potential “stop-and-go” traffic pattern. One-minute intervals were considered sufficient for this study area. Shorter intervals might be more appropriate in a densely settled region and could easily be entered as inputs to the model. The primary travel time (TPT) for a given vehicle is determined when the vehicle leaves the evacuation zone. These operations in modules 2, 3, and 4 are repeated for all primary roadways. As with the NETVACI model (see note 8), this model could also be modified to include random interruptions (accidents or disabled vehicles) or the effects of traffic control.

Statistical Analysis

Repeated Monte Carlo trials were performed as indicated in Figure 3 to form a sufficient statistical base for estimating evacuation time. The computer model simulates evacuation by using 25 different sets of random mobilization times in order to provide a range of time estimates, as well as an average expected result and normal variations (standard deviations). Twenty-five replications were judged to provide a sufficiently large sample for purposes of statistical analysis. The number of runs could be increased arbitrarily if desired.

Module 5 performs these statistical analysis calculations. Since in reality there is no way to predict who will be on the road in 15 minutes and who will take an hour or

longer, this approach allows for the possibility that those who are slow to pack up could be located anywhere within the plume exposure EPZ. By using 25 separate and different simulations, the probability of being misled by a single, unusually rapid (or unusually slow) evacuation is greatly reduced.

SAMPLE INTERPRETATIONS

Outputs from the computer model include an estimate of the average evacuation time (excluding notification and confirmation time) as shown in Table 2 for the 1980 population. Estimates of the corresponding standard deviations are given in parentheses. All ten evacuation areas are considered in Table 2 with both normal and adverse weather conditions and the four "incident time of occurrence" cases, a total of 88 separate simulations, each involving 25 runs with different random mobilization times.

Also provided as output from the computer model are tables that show the detailed progress of the evacuation every 5 minutes by major exit routes and nodes. For example, Table 3 shows the details for one representative evacuation of the total 1980 nighttime population under normal weather conditions. Table 2 shows that for 25 simulation runs the 4,510 vehicles were evacuated in an average time of 143 minutes with a standard deviation of 8 minutes.

Assuming the results can be approximated by a normal distribution implies that the evacuation would be completed within 159 minutes 97.5% of the time (plus two standard deviations); and that the evacuation time would exceed 167 minutes (plus three standard deviations) only once in approximately 200 runs. The 25 simulations yielded an evacuation time ranging from a low of 127 minutes to a maximum of 160 minutes. Thus, the output from the model gives decision makers a range of time estimates using a variety of assumptions, allowing the flexibility of planning for either average or worst-case situations.

The model can also be used to identify specific problem areas. When traffic congestion occurs, the output lists the node involved and the number of vehicles delayed. Two types of traffic problems are monitored. First, a node may become a bottleneck causing traffic to back up along a major exit route (nodes 24 and 52 on 71st Street near Broken Arrow could be examples of this problem). Second, a major route may be operating at or near capacity, resulting in a delay for vehicles attempting to enter the evacuation network at nodes along that portion of the route (i.e., an entry queue forms). Nodes 52 and 53 on 71st Street encounter this problem.

In spite of the predicted traffic congestion on 71st Street, State Highway 33 heading toward Tulsa was the last route to clear for 17 of the 25 trials. Due to the Inola population, the larger number of vehicles assigned to that route increases the probability that one of those vehicles will be among the last to mobilize, start on the road, encounter road congestion, and, thus, the last to exit.

In this specific case study, mobilization time was found to be the major component of total evacuation time. For most vehicles, travel time is relatively brief. While 99% of the population is assumed to be ready to move in 85 minutes, the achievement of 99% evacuation occurs about 20 minutes later at the 105-minute mark. On the average, an additional 39 minutes are required for the evacuation of stragglers, i.e., those with the largest mobilization times. Finally, adverse weather conditions increased evacuation time estimates in most cases by about 20% (20–25 minutes).

Table 2. A Summary of Computer Model Predictions for the Total Evacuation of the 1980 Population (Evacuation Time in Minutes).

Area Evacuated*	Nighttime					Worktime				
	Vehicles	Normal		Adverse		Vehicles	Normal		Adverse	
NE2	73	121 (9)	146	(9)		30	109 (11)	134	(11)	
SW2	61	117 (9)	142	(9)		27	109 (12)	133	(12)	
N5	1201	140 (8)	163	(9)		761	134 (9)	156	(7)	
E5	369	130 (12)	154	(12)		158	124 (12)	146	(12)	
S5	190	126 (11)	151	(11)		78	118 (9)	143	(9)	
W5	198	124 (11)	148	(12)		116	117 (12)	139	(13)	
N10	1601	140 (8)	162	(8)		964	135 (8)	157	(8)	
E10	582	133 (10)	155	(11)		229	125 (13)	148	(14)	
S10	684	129 (11)	151	(11)		277	123 (15)	146	(16)	
W10	2219	134 (8)	211	(1)		983	127 (10)	142	(13)	
Total	4510	143 (8)	220	(1)		2175	139 (8)	159	(9)	

Area Evacuated*	Sunday Morning					Summer Saturday				
	Vehicles	Normal		Adverse		Vehicles	Normal		Adverse	
NE2	80	119 (10)	145	(11)		114 (11)	140 (11)			
SW2	69	119 (9)	143	(10)		235	130 (13)	154	(13)	
N5	1235	141 (9)	164	(9)		1040	140 (8)	163	(8)	
E5	358	129 (11)	152	(11)		450	134 (11)	158	(12)	
S5	217	131 (12)	155	(12)		521	136 (10)	161	(11)	
W5	205	126 (13)	149	(14)		369	130 (11)	154	(11)	
N10	1617	140 (8)	162	(7)		1320	141 (8)	164	(7)	
E10	568	132 (11)	153	(11)		613	134 (12)	159	(11)	
S10	659	129 (10)	152	(11)		865	137 (9)	159	(10)	
W10	1992	133 (9)	183	(1)		1779	132 (7)	154	(6)	
Total	4264	141 (8)	192	(1)		3596	144 (9)	166	(9)	

Note: *The notation N10 refers to evacuation of the north sector out to the 10-mile radius; N5 would be included within N10. Sums of partial evacuation exceed the total due to double-counting along sector boundaries.

CONFIRMATION TIME ESTIMATES

Confirmation time estimates can be based both on results of the statistical time analysis of Module 5 and the particular assignments of evacuation confirmation teams to given areas within the plume exposure EPZ boundary. The number of confirmation team personnel to be assigned to each area depends upon the relative estimated evacuation

Table 3. Computer Model Output Showing the Average Exiting Time for the Total 1980 Nighttime Population.

Time Elapsed (Minutes)	Number of Cars Exited by Route or Node									Total Cars Exited	Cars Exited (%)
	18	21	4	25	88N	33E	715T	33W	Other		
5	0	2	1	2	3	0	0	0	2	10	0.22
10	0	3	5	6	7	2	21	4	13	61	1.35
15	0	5	7	10	14	2	76	20	27	171	3.79
20	3	11	14	20	34	8	145	69	49	353	7.83
25	9	13	20	26	64	18	216	127	69	562	12.46
30	18	20	29	31	89	39	286	230	103	845	18.74
35	33	23	39	41	122	77	358	338	145	1176	26.08
40	53	26	51	45	159	117	433	485	183	1552	34.41
45	82	26	61	55	196	150	506	633	232	1941	43.04
50	101	28	66	59	255	192	580	767	278	2326	51.57
55	118	31	80	68	295	336	651	904	316	2688	59.60
60	142	31	84	76	332	225	719	1033	339	3011	66.76
65	167	32	89	82	358	275	792	1139	364	3298	73.13
70	180	34	93	87	372	299	867	1227	384	3543	78.56
75	197	34	100	90	384	315	936	1294	393	3743	82.99
80	206	37	102	90	392	328	1001	1366	408	3930	87.14
85	216	38	102	93	401	337	1056	1411	417	4071	90.27
90	229	39	102	95	408	345	1108	1459	422	4207	93.28
95	240	39	103	95	411	351	1171	1485	424	4319	95.76
100	246	39	103	95	411	356	1219	1503	427	4399	97.54
105	248	39	103	95	412	357	1266	1514	431	4465	99.00
110	251	40	103	95	413	357	1269	1520	431	4479	99.31
115	253	40	103	95	414	357	1270	1527	431	4490	99.56
120	254	40	103	95	415	357	1270	1531	431	4496	99.69
125	255	40	104	95	415	357	1270	1533	431	4500	99.78
130	255	40	104	95	416	357	1271	1536	431	4505	99.89
135	256	40	104	95	416	357	1272	1536	431	4507	99.93
140	256	40	104	95	416	358	1272	1537	431	4509	99.98
144	256	40	104	95	416	358	1272	1537	432	4510	100.00

time, the population density, and the total number of miles of local access roads within that area. These teams would begin their confirmation tasks even before the last evacuees have left the area. For example, confirmation within an area might begin when 99% of the population has been evacuated, which can be estimated from the output of Module 5 of the model. The remaining 1% can be identified by the confirmation team, and their evacuation can be verified when they arrive at a particular collector node for entry into a primary road. In the case shown in Table 3, confirmation could begin at the 105-minute mark. In any of the low population sectors, confirmation could reasonably be started even earlier.

CONCLUSIONS

A methodology has been described for estimating the emergency evacuation times of a partial or total EPZ. This methodology involves a computer model that requires as

inputs mobilization time curves and details of roadways forming an evacuation network. Outputs of the computer model provide information on average evacuation times with standard deviations for the 10 (partial or total) evacuation areas with normal or adverse weather and 4 ITO cases. Bottlenecks can be identified from the computer model outputs to aid in implementing reassignment of vehicle routes and/or improved traffic control during evacuation.

The time required for total evacuation depends on assumptions about the time of departure and movement of the last vehicle to "clear" the plume exposure EPZ boundary. However, the evacuation model can provide estimates of the number (or percentage of the total number) of vehicles that have "cleared" at any of a varied selection of time intervals after the evacuation order has been given—30 minutes after, 45 minutes, 1 hour, 80 minutes, 3 hours, or other periods of elapsed time.

One general assumption is that mobilization time of individual household units can be adequately described by a probability density function. Traffic capacities (vehicles per hour) and node segment densities (vehicles per mile) are used as model inputs to predict traffic movement. Another assumption is that the use of repeated computer runs (Monte Carlo trials) provides a statistical base that can be analyzed in order to assess expected model accuracy.

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